


CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

Test Pattern

JEE(Advanced)
FULL SYLLABUS
 17-04-2024

JEE(Main + Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

Time : 3 Hours

PAPER-2 (OPTIONAL)

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY
GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **20** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

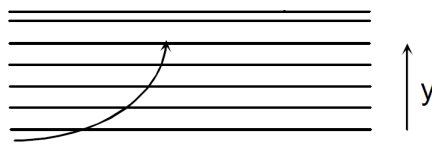
Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

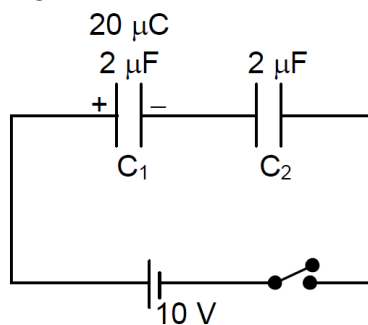
1. The refractive index of the medium within a certain region, $x > 0$, $y > 0$, changes with y . A thin light ray travelling in the x -direction strikes the medium at right angles and moves through the medium along a circular arc. How does the refractive index depend on y ?

[R : Radius of path & η_0 : Initial refractive index]



- (A) $\frac{2R}{R-y} \eta_0$
- (B) $\frac{R}{R-y} \frac{\eta_0}{2}$
- (C) $\frac{R+y}{R} \eta_0$
- (D) $\frac{R}{R-y} \eta_0$
2. A mass is suspended from a wire and is pulled along the length of wire, resulting in oscillations of time period T_1 . The same mass is next attached to wire of the same material and length but double the cross-sectional area. The time period this time is T_2 . Then T_1/T_2 is equal to
- (A) 1 : 2
- (B) 2 : 1
- (C) 1 : $\sqrt{2}$
- (D) $\sqrt{2}$: 1

3. Initially, C_1 is charged but C_2 is uncharged. When switch is closed, charges on C_1 and C_2 are



- (A) $20\ \mu\text{C}$, 0
 (B) $10\ \mu\text{C}$, $10\ \mu\text{C}$
 (C) $30\ \mu\text{C}$, $10\ \mu\text{C}$
 (D) 0 , $20\ \mu\text{C}$
4. A standing wave is produced in a vapour of an element of atomic mass $254\ (\text{g mol}^{-1})$ at $400\ \text{K}$. Nodes are found to be $6.77\ \text{cm}$ apart when frequency of source was $1000\ \text{Hz}$. The vapour used for the experiment is ($R = 8.31\ \text{JK}^{-1}\text{mol}^{-1}$)
- (A) monatomic
 (B) diatomic
 (C) polyatomic
 (D) none of these

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. A particle of mass m is moving in a field where the potential energy is given by $U(x) = U_0 (1 - \cos ax)$, where U_0 and a are constants and x is the displacement from mean position. Then (for small oscillation)

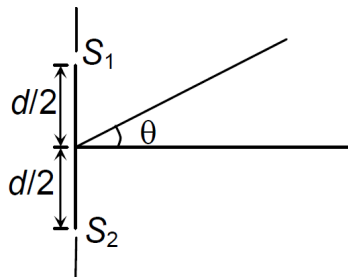
(A) the time period is $T = 2\pi \sqrt{\frac{m}{aU_0}}$

(B) the speed of particle is maximum at $x = 0$

(C) the amplitude of oscillations is $\frac{\pi}{a}$

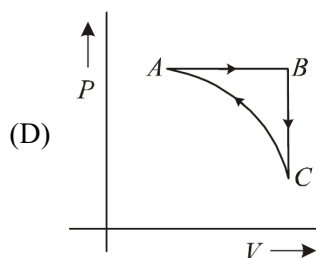
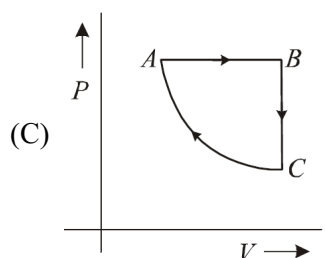
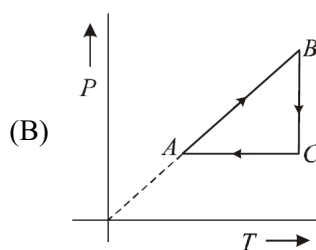
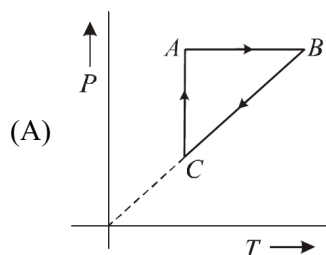
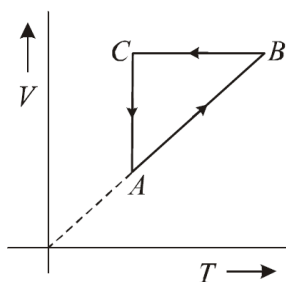
(D) the time period is $T = 2\pi \sqrt{\frac{m}{a^2 U_0}}$

6. In an interference arrangement similar to Young's double slit experiment, the slits S_1 and S_2 are illuminated with coherent microwave sources, each of frequency 10^6 Hz . The sources are synchronized to have zero phase difference. The slits are separated by a distance $d = 150.0 \text{ m}$. The intensity $I(\theta)$ is measured as a function of θ , where θ is defined as shown. If I_0 is maximum intensity, then $I(\theta)$ for $0 \leq \theta \leq 90^\circ$ is given by

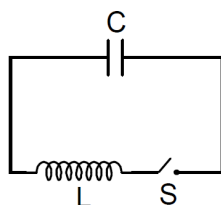


- (A) $I(\theta) = \frac{I_0}{2}$ for $\theta = 30^\circ$
- (B) $I(\theta) = \frac{I_0}{4}$ for $\theta = 90^\circ$
- (C) $I(\theta) = I_0$ for $\theta = 0^\circ$
- (D) $I(\theta)$ is constant for all values of θ
7. A uniform square plate of mass m and edge a initially at rest starts rotating about one of the edge under the action of a constant torque τ . Then at the end of the 5^{th} sec after start
- (A) angular momentum is equal to 5τ
- (B) kinetic energy is equal to $\frac{75\tau^2}{ma^2}$
- (C) angular momentum is equal to 2.5τ
- (D) kinetic energy is equal to $\frac{75\tau^2}{2ma^2}$
8. A particle of mass m start moving at $t = 0$ due to a force $F = F_0 \cos \omega t$, where F_0 and ω are constant. Particle will stop first time at $t = t_0$ and distance travel in this time interval is S , then
- (A) $T_0 = \frac{\pi}{\omega}$
- (B) $t_0 = \frac{\pi}{2\omega}$
- (C) $S = \frac{2F_0}{m\omega^2}$
- (D) $S = \frac{F_0}{m\omega^2}$

9. An ideal gas undergoes a thermodynamic cycle as shown in figure. Which of the following graphs represents the same cycle ?



10. In the given circuit, initially the charge on capacitor is Q_0 . At $t = 0$, the switch S is closed. Which of the following statement(s) is/are correct ?



- (A) Maximum current through the inductor is $\frac{Q_0}{\sqrt{LC}}$.
- (B) Charge on the capacitor is zero at $t = \frac{\pi}{2}\sqrt{LC}$.
- (C) Current through the inductor is maximum at $t = \frac{\pi}{2}\sqrt{LC}$.
- (D) Maximum magnetic energy can be stored in the inductor is $\frac{Q_0^2}{2C}$.

Note : No questions have been asked in the section II of this paper. Please leave section II blank on the OMR answer sheet.

SECTION-III : (Maximum Marks: 24)

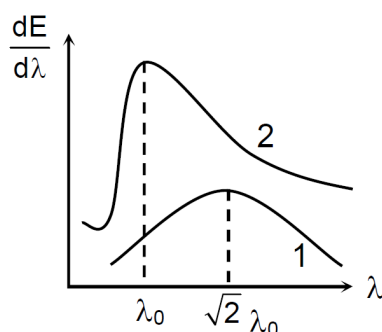
- This section contains **EIGHT (08)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.

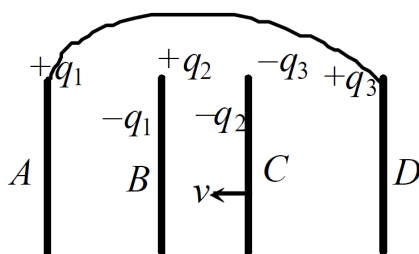
Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

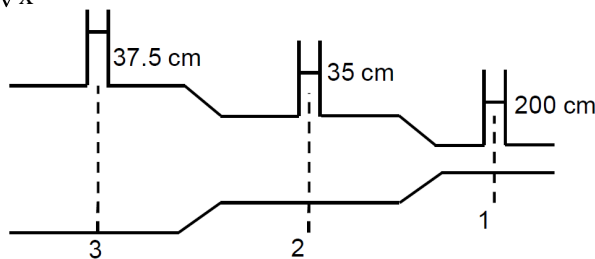
1. A vertical hollow cylinder contains an ideal gas. The gas is enclosed by a 5 kg movable piston with area of cross-section $5 \times 10^{-3} \text{ m}^2$. The gas is slowly heated from 300 K to 350 K and the piston rises by 0.1 m. The piston is now clamped at this position and the gas is cooled back to 300 K. The difference between the heat energy added during heating process and heat energy lost during cooling process is $11x$. Find value of x . [Take $p_0 = 10^5 \text{ N/m}^2$]
2. Graph 1 and 2 represent the spectral emissive power of a body at temperature T_1 and T_1 . The ratio of area under curve 2 and curve 1 is.



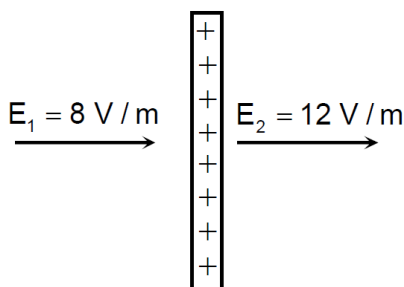
3. In the diagram there are four conducting plates A, B, C and D placed parallel to each other at equal separation L . If plate C starts moving towards plate B with velocity v . Find the current (in mA) flowing in the wire connecting A and D. (assume all other plates to be fixed)
(Given: $q_2 = 2\mu\text{C}$, $q_3 = 3\mu\text{C}$, $v = 30 \text{ m/s}$, $L = 0.05 \text{ m}$)



4. Water is flowing in varying cross-section pipe. The areas of cross-sections 1, 2 and 3 are 1 cm^2 , 2 cm^2 and $A \text{ cm}^2$, respectively. Water levels are shown in different vertical tubes of manometers. The speed of water at cross-section 3 is $\frac{1}{\sqrt{x}} \text{ m/s}$. The value of x is



5. If the K_{α} radiation of Mo ($Z = 42$) has a wavelength of $0.71 \text{ \AA}$, calculate wavelength of the corresponding radiation of Cu. i.e. K_{α} for Cu ($Z = 29$). Find the value of nearest integer value (in $\text{\AA}$) of wavelength.
6. A particle is suspended by a light vertical inelastic string of length ℓ from a fixed support. At its equilibrium position, it is projected horizontally with a speed $\sqrt{6g\ell}$. What is the ratio of the tension in the string in its horizontal position to that in string, when the particle is vertically above the point of support?
7. The electric field on two sides of a charged plate is shown in the figure. If the charge density on the plate is $n\epsilon_0$, then n is



8. A hydrogen like atom (atomic number z) is in a higher excited state of quantum number n . The excited state by successively emitting two photons of energy 10.2 eV and 17.0 eV , respectively. Alternatively, the atom from the same excited by successively emitting two photons of energies 4.25 eV and 5.95 eV , respectively. Determine the value of n . (Take, ionisation energy of H-atom = 13.6 eV)

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

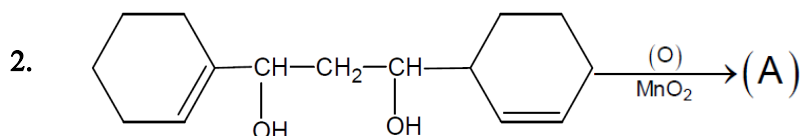
Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

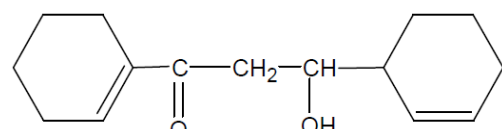
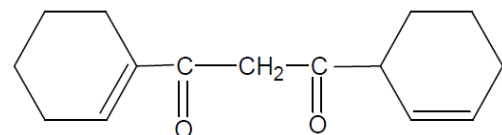
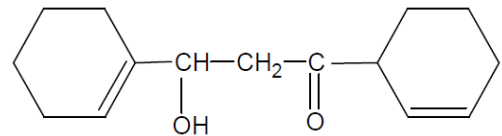
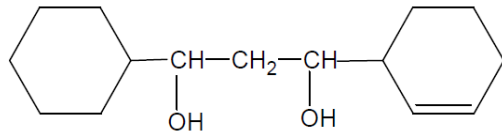
Negative Marks : -1 In all other cases

1. For $[\text{HgI}_3]^-$ ion geometry and magnetic behaviour is respectively:

- (A) triangular bipyramidal, paramagnetic
(B) triangular pyramidal, diamagnetic
(C) triangular planar, paramagnetic
(D) triangular planar, diamagnetic



The product (A) is

- (A) 
(B) 
(C) 
(D) 

3. For most reactions, rate of a reaction increases about two times when its temperature is raised by 10 K. This fact is due to
- (A) increase in number of collisions per unit volume
 - (B) increase in number of molecules having energy E_A or more
 - (C) lowering of activation energy
 - (D) lowering of pre-exponential factor in Arrhenius equation
4. At constant volume, for a fixed amount of a gas the pressure of the gas increase with rise of temperature due to
- (A) increase in average molecular speed
 - (B) increased rate of collision amongst molecules
 - (C) increase in molecular attraction
 - (D) decrease in mean free path

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

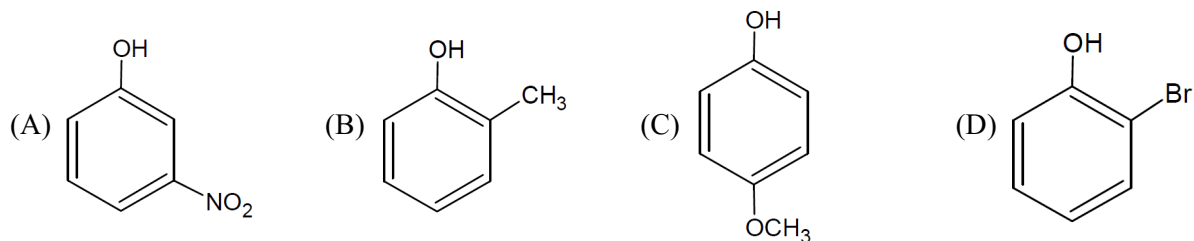
5. Which of the following statements are correct ?

- (A) Copper metal can reduce Fe^{2+} in acidic medium.
- (B) The absolute value of standard electrode potential can be determined experimentally
- (C) In a Daniell cell, electrons flow from zinc electrode to copper electrode outside the cell
- (D) Cell voltage is an intensive quantity.

6. Which of the following show the correct order of nucleophilicity ?

- (A) $\text{CH}_3\text{S}^- > \text{CH}_3\text{O}^-$
- (B) $(\text{CH}_3)_3\text{N} > (\text{CH}_3)_3\ddot{\text{P}}$
- (C) $\text{CN}^- > \text{Cl}^-$
- (D) $\text{OC}_2\text{H}_5^- > \text{OH}^-$

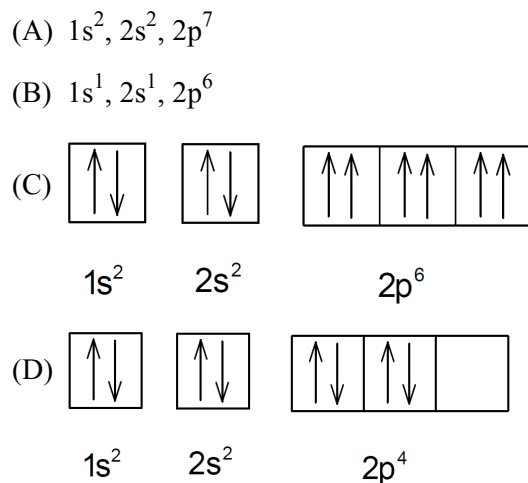
7. Which of the following is/are more acidic than phenol ?



8. Select molecule(s) in which all the π -orbitals are in same plane.

- (A) C_6H_6
 (B) $O = C = C = C = O$
 (C) $H_2C = C = CH_2$
 (D) $CH_2 = CH - CH = CH_2$

9. In which of the following Pauli rule is violated ?



10. Synergic bonding in metal carbonyl complexes $[M = \text{Metal}]$:

- (A) decreases C – O bond strength
 (B) increases M – C bond length
 (C) increases bond order for M – C bond
 (D) increases paramagnetism of complex

Note : No questions have been asked in the section II of this paper. Please leave section II blank on the OMR answer sheet.

SECTION-III : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.

Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

-
1. Find the total number of electron in Cr which have $(n \times l) = \text{zero}$
 2. Total number of exchanges possible for $3d^4$ electronic configuration are?
 3. Total number of bonds in PCl_5 which are at 90° to each other, are
 4. Consider octahedral complex $\text{K}_2[\text{Ni}(\text{EDTA})]$.
Total no. of 5 member rings in compound = X
Total number of geometrical isomers possible for compound = Y
Represent your answer as a total of X and Y.
 5. Consider a titration of potassium dichromate solution with acidified Mohr's salt solution using diphenylamine as indicator. The amount of Mohr's salt required per mole of dichromate is
 6. The number of atoms in a single face-centred cubic unit cell is
 7. Total no. of geometrical isomers possible for 1, 2, 4-tribromocyclohexane are?
 8. Total no. of stereoisomers possible for 1, 2-dimethylcyclopropane are?

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. If the area bounded by $f(x) = \sqrt{1+x^2}$ ($x > 0$) the line $y = x$, y axis and $y = -x + a$ ($a > 1$) is k , then area bounded by the graph of $f^{-1}(x)$, the line $y = x$ between the lines $y = -x + 1$ and $y = -x + a$ is

(A) $\frac{k-1}{4}$	(B) $\frac{2k-1}{2}$
(C) k	(D) $\frac{4k-1}{4}$
2. In a acute angle triangle ABC, the foot of the perpendicular from A divide the opposite side into two parts of length 3 and 17 units and $\tan A = \frac{22}{7}$, the $\left(\frac{\text{Area of } \triangle ABC}{22} \right)$ is

(A) $\frac{5}{2}$	(B) 5
(C) 10	(D) 20
3. The digits of three digit number N are in A.P. If sum of the digits is 15 and the number obtained by reversing the digits of the number is 594 less the original number, then $\frac{1000}{N-252}$ is equal to :

(A) $\frac{5}{6}$	(B) $\frac{5}{3}$
(C) 0.06	(D) 0.03
4. A flagstaff stands vertically on a pillar, the height of the flagstaff being double the height of the pillar. A man on the ground at a distance finds that both the pillar and the flagstaff subtend equal angles at his eyes. The ratio of the height of the pillar and the distance of the man from the pillar, is

(A) $\sqrt{3} : 1$	(B) 1 : 3
(C) 1 : $\sqrt{3}$	(D) $\sqrt{3} : 2$

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. Let $\vec{a}, \vec{b}, \vec{c}$ be non-coplanar vectors and $\vec{p} = \frac{\vec{b} \times \vec{c}}{[\vec{a} \vec{b} \vec{c}]}, \vec{q} = \frac{\vec{c} \times \vec{a}}{[\vec{b} \vec{c} \vec{a}]}, \vec{r} = \frac{\vec{a} \times \vec{b}}{[\vec{c} \vec{a} \vec{b}]}$ then

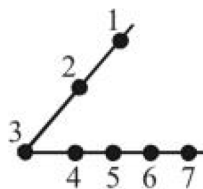
(A) $\vec{p} \cdot \vec{a} = 1$

(B) $\vec{p} \cdot \vec{a} + \vec{q} \cdot \vec{b} + \vec{r} \cdot \vec{c} = 3$

(C) $\vec{p} \cdot \vec{a} + \vec{q} \cdot \vec{b} + \vec{r} \cdot \vec{c} = 0$

(D) none of these

6. 7 points are given in a plane (as shown in adjacent figure), then using these 7 points –



(A) Number of lines that can be formed is 10.

(B) Number of triangles that can be formed is 24.

(C) Number of quadrilateral that can be formed is 6.

(D) Maximum number of circles that can be formed is 24, where each circle contains atleast three of 7 points

7. The equations of four circles are $(x \pm a)^2 + (y \pm a)^2 = a^2$. The radius of a circle touching all the four circles is
- (A) $(\sqrt{2} - 1)a$
(B) $2\sqrt{2}a$
(C) $(\sqrt{2} + 1)a$
(D) $(2 + \sqrt{2})a$
8. If $f(x)$ is a cubic polynomial such that $f(0) + 2 = 0$ and $f'''(0) = 6$ & $f(x) = 0$ possesses three positive integral roots then
- (A) $f'(1) = 0$
(B) $2f'(0) + f''(0) = 0$
(C) $2f'(0) + f''(0) = 2$
(D) $f'(2) = 0$
9. If $x + [y] + \{z\} = 1.1$ $[\cdot]$ is G.I.F. and $\{\cdot\}$ is fractional part
 $[x] + \{y\} + z = 2.2$
 $\{x\} + y + [z] = 3.3$ then
- (A) $x + y + z = 3.3$
(B) $y - 2x = 1$
(C) $2(z + 1) = 5y$
(D) $\{x\} + \{y\} + \{z\} = 0.3$
10. If the sides a, b & c of a triangle ABC are in G.P. then $\angle B$ can never be
- (A) $\frac{\pi}{6}$
(B) $\frac{5\pi}{6}$
(C) $\frac{\pi}{4}$
(D) $\frac{2\pi}{3}$

Note : No questions have been asked in the section II of this paper. Please leave section II blank on the OMR answer sheet.

SECTION-III : (Maximum Marks: 24)

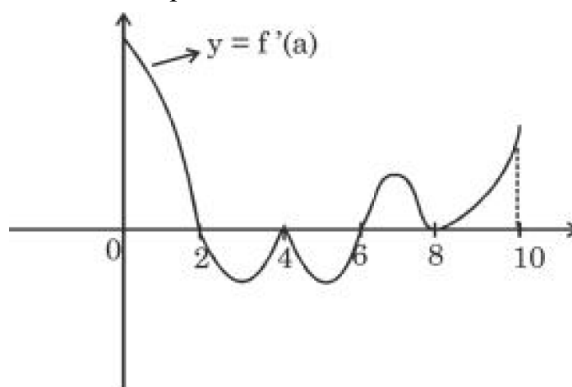
- This section contains **EIGHT (08)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.

Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

1. If $S = 3.\sqrt{9}.\sqrt{\sqrt{27}}.\sqrt{\sqrt{\sqrt{81}}}\dots\infty$, then sum of digits in S is
2. Let $f(x) = (1+x)\ln(1+x) - x - \frac{x^2}{4}$, $\forall x > -1$ and $f'(x) > 0 \forall x \in (0, \alpha)$, then value of $[\alpha]$ is (where $[\]$ denotes greatest integer function)
3. Let $f(x)$ be a derivable function such that $f(0) = 1$, $f(2) = 2$, $f(4) = 1$ and $f(6) = -4$. The graph of $y = f'(x)$ is given below. Then find the total number of points of local maximum/local minimum of $y = f(x)$ in $(0, 10)$



4. The projection length of a variable vector $x\hat{i} + y\hat{j} + z\hat{k}$ on the vector $\vec{p} = \hat{i} + 2\hat{j} + 3\hat{k}$ is 6. Let ℓ be the minimum projection length of the vector $x^2\hat{i} + y^2\hat{j} + z^2\hat{k}$ on the vector $\vec{p}$, then the value of $\sqrt[3]{\ell^2 + 15^2}$ is
5. Non-zero vectors $\vec{a}, \vec{b}, \vec{c}$ satisfy $\vec{a} \cdot \vec{b} = 0$, $(\vec{b} - \vec{a}) \cdot (\vec{b} + \vec{c}) = 0$ and $2|\vec{b} + \vec{c}| = |\vec{b} - \vec{a}|$. If $\vec{a} = \mu\vec{b} + 4\vec{c}$ then the value of μ is
6. 3 couples have to be seated around a circle. Let p be the probability that no couple is together then the value of $30p$ is
7. If $f(0) = 1$, $f(2) = 3$, $f'(2) = 5$ then the value of the definite integral $\int_0^1 xf''(2x)dx$ is
8. A special die is so constructed that the probabilities of throwing 1, 2, 3, 4, 5 and 6 are $(1-k)/6$, $(1+2k)/6$, $(1-k)/6$, $(1+k)/6$, $(1-2k)/6$ and $(1+k)/6$ respectively. If two such dice are thrown and the probability of getting a sum equal to 9 lies in $\left[\frac{1}{9}, \frac{2}{9}\right]$. Then find the number of integral solutions of k .

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work

**CALL teleMANAS**

Toll Free No.

☎ 14416, 1800-8914416

ALLEN De-Stress No.

☎ 0744-2757677 📞 +91-8306998982

ALLEN CAREER INSTITUTE Pvt. Ltd.

Registered & Corporate Office : 'SANKALP', CP-6, Indra Vihar, Kota (Rajasthan) INDIA-324005

Ph. : +91-744-3556677, +91-744-2757575 | E-mail : info@allen.in | Website : www.allen.ac.in

E-20/20

Your Target is to secure Good Rank in JEE 2024

1001CJA106216230183